INDIA STRATEGY

Room To Rebound

Amorita Goel, CFA
+65-6654-5445 | amorita.goel@gs.com
Goldman Sachs (Singapore) Pte

Sunil Koul
+44(20)7051-4931 | sunil.koul@gs.com
Goldman Sachs International

Timothy Moe, CFA
+65-6889-1199 | timothy.moe@gs.com
Goldman Sachs (Singapore) Pte

Room To Recover. India's outlook has improved in recent weeks, amid lower commodity prices, stabilized currency, resilient domestic growth, healthy earnings expectations in 2Q, and potential recovery in select domestic pockets. With ultra-light foreign positioning, we see ample room for flows to return. While renewed geopolitical tensions in the Middle East may fuel near-term volatility, we see room for NIFTY to recover towards our June 2027 target of 26,500, implying c.10% upside from current levels following a 9% drawdown in 1H. For investors looking to re-engage in Indian equities, we identify key macro themes at play and highlight our top alpha picks for 2H.

Expect Rotation To Value, Large-caps And Most Sold Foreign Pockets: Valuation de-rating drove negative market returns in 1H on fears of economic slowdown, while 'Growth' outperformed 'Value' amid scarcity of earnings. Looking ahead, we expect a rotation from 'Growth' to 'Value' as investors look for reasonably valued pockets in anticipation of a recovery. As foreign outflows reverse course through 2H, most sold and reasonably valued pockets like Large-caps and Banks stand to gain the most.

Key Events To Monitor: 2Q Earnings, Bond Inflows, And 'Super' El Niño. As Q2 2026 results roll in, dislocations between expected and actual oil-shock impacts on profits will yield tactical opportunities. Analysts have likely over-cut estimates for energy refiners, where tight capacity protects margins, and tourism/airline companies, where demand has been more resilient and oil costs have moderated. Beyond energy, strong bond inflows in 2H and a steady INR will tilt the scales toward domestic sectors over exporters. Meanwhile, a looming 'super' El Niño presents a dual narrative: a headwind for the wider market and a drag on rural-focused stocks, but a catalyst for power utilities facing supply shortages.

Trade Ideas For 2H: Thematically, we recommend going long on banks, tourism and energy refiners, and favor Large-caps over Mid-caps, value over growth, power utilities over rural/agriculture stocks, and domestics over exporters. **Structurally,** we remain bullish on defense and energy security. **Sectorally**, we raise utilities to overweight and remain overweight on banks, energy refiners, TMT, and defense. We stay underweight exporters, downstream oil and downgrade select materials to underweight. For **stock picks**, we highlight 15 names poised to benefit from near-term catalysts and medium-term structural themes.

1. Improved Macro Outlook Amid Domestic Resilience

India's macro outlook has improved in recent weeks, amid lower commodity prices, stabilized currency, resilient domestic growth, expectation of healthy profit delivery in 2Q, and potential EPS upgrades in select domestic pockets despite a broader earnings downcycle. After being hit by the longest supply-driven oil rally in the last 15 years, and a sharp currency depreciation, India's current EPS downgrade cycle is on track to be the weakest in recent history. Consensus has cut MSCI India's CY26e earnings growth by 4pp to 12%, coming closer towards our full-year estimate of 10%. The cuts could further deepen in the next couple of months, per historical precedents, as analysts bake in further impact of higher oil prices and weaker currency on corporate profits with a lag. However, under the hood, analysts may have over-cut estimates on oil shock-related impact in pockets of energy, consumer retail/services, chemicals/cement and staples. On the other hand, cuts may have been understated in metals/mining and healthcare, per past evidence. As we enter 2Q 2026 results season, analysts are estimating another healthy quarter with 12% yoy profit growth for MSCI India ex-commodities, despite the quarter falling in the middle of the energy crisis, reflecting India's domestic resilience. The dislocations between expected and actual oil-shock impacts on profits will yield tactical opportunities for investors to trade.

Exhibit 1: India's macro outlook has improved in recent weeks, albeit still weaker than pre-war months

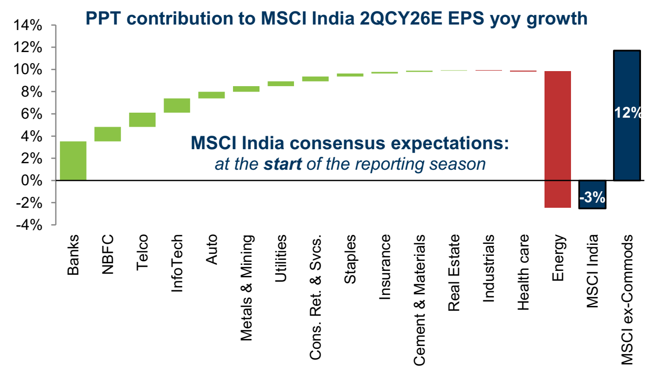
India's GS Macro forecasts for CY26 Pre- and Post US-Iran War

	Pre War Forecast	Post War Forecast	Post US-Iran Deal Forecast
Real GDP	7.1	5.9	6.8
CPI Inflation (average)	3.9	4.6	4.4
Current account balance (% of GDP)	-0.9	-2.0	-1.1
Repo rate (%, eop)	5.25	5.75	5.75
Fiscal Balance (% of GDP)	-4.3	-4.7	-4.7
Brent average price ($/bbl)	64	95*	85
12m USDINR	94	97	96

**We assumed Indian oil importers paid a premium of c.$10/bbl above $85/bbl Brent price in our 26 March update. We remove this assumption in our 19 June update.*

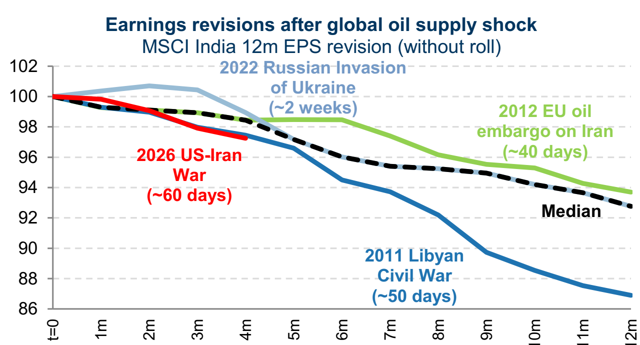
Source: Goldman Sachs Global Investment Research

Exhibit 2: Consensus is expecting healthy 12% yoy profit growth for MSCI India ex-commodities in 2Q CY26E



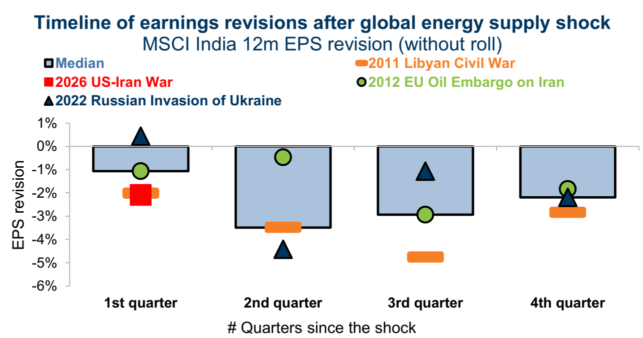
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 3: Current EPS downgrade cycle in India is on track to be among the weakest in history



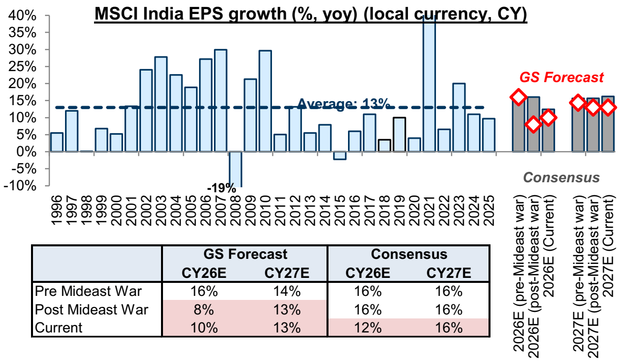
Source: Goldman Sachs Global Investment Research

Exhibit 4: Historical trends suggest earnings cuts deepen in the 2nd quarter post the oil supply shock



Source: Goldman Sachs Global Investment Research

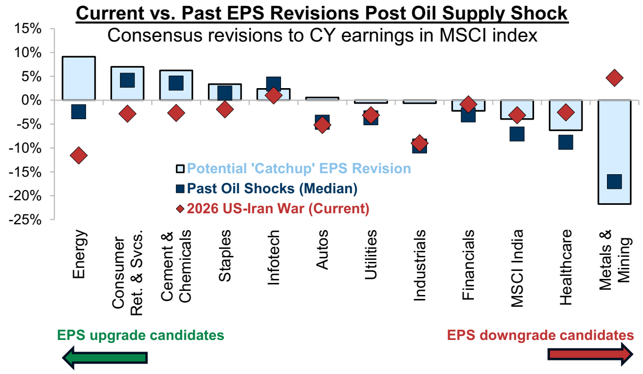
Exhibit 5: We expect consensus to cut earnings growth further by 2pp to 10% for CY26 (from 12%)



	GS Forecast		Consensus	
	CY26E	CY27E	CY26E	CY27E
Pre Mideast War	16%	14%	16%	16%
Post Mideast War	8%	13%	16%	16%
Current	10%	13%	12%	16%

Source: Goldman Sachs Global Investment Research

Exhibit 6: Comparing 2026 oil shock to past episodes, recent EPS cuts may have been overdone in energy and consumer retail/services, while they may be understated in metals/mining and healthcare

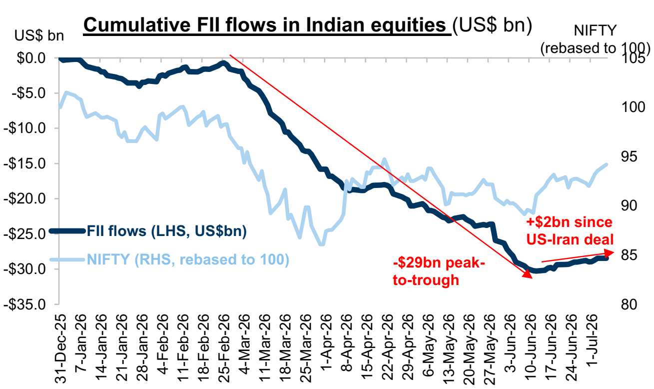


Source: FactSet, Goldman Sachs Global Investment Research

2. Foreign Inflows Should Gradually Return

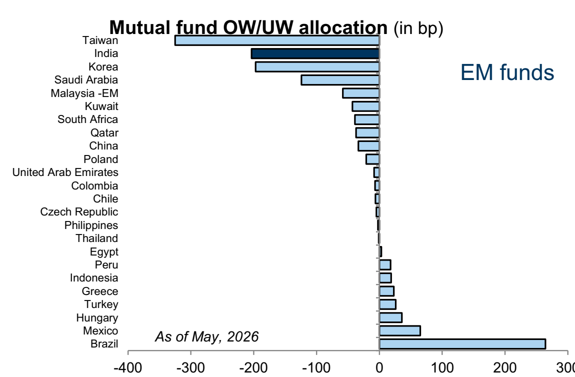
Foreign selling is likely over, and sentiment should turn incrementally favorable on improved domestic outlook and ultra-light foreign positioning. Global equity investors used India as a funding market through 1H 2026, selling a record ~US$30bn of Indian equities in a short span of 3.5 months. They have turned net buyers since mid-June, albeit modestly so, with $2bn inflows, largely in Financials. With large underweight positioning towards Indian equities, global funds have ample room to neutralize their exposure. While a continued earnings downgrade cycle and still less attractive growth-valuation mix relative to other markets will be key investor concerns, improving visibility on domestic recovery will act as a catalyst for investors to start pricing in the anticipated recovery in advance.

Exhibit 7: Foreign flows in Indian equities have turned modestly positive since mid-June



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 8: India is one of the most underweight markets in EM funds, suggesting ample room to neutralize exposure to India

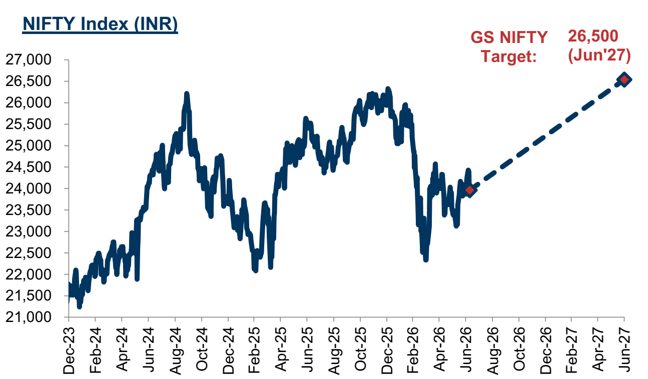


Source: EPFR, Goldman Sachs Global Investment Research

3. Room To Rebound

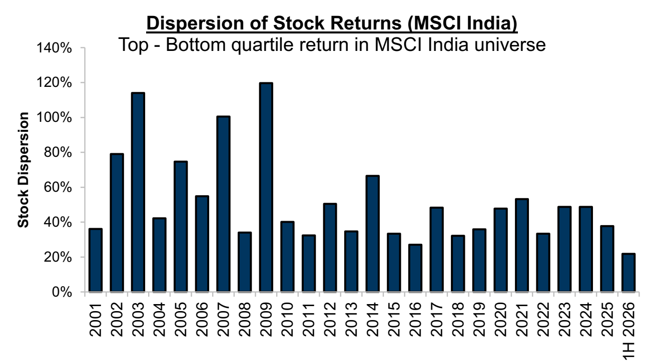
After a large equity drawdown in 1H, we see room to rebound from the lows. NIFTY lost 9% in INR in 1H 2026, and relative to the regional markets, Indian equities saw their worst returns in 3 decades. Looking ahead, while renewed geopolitical tensions in the Middle East may fuel near-term volatility, as domestic recovery starts to get priced in, we expect NIFTY to rebound towards our June'27 target of 26,500, implying c.10% upside from current levels. As the broad-market re-gains its momentum, we identify key macro themes at play and highlight our top alpha picks for investors looking to re-engage in India. Intra-market alpha opportunities could grow as stock dispersion improves from current lows.

Exhibit 9: After a 9% drawdown in NIFTY in 1H, we see room to recover from here on, towards our June'27 target of 26,500, implying c.10% upside



Source: Goldman Sachs Global Investment Research

Exhibit 10: Intra-market alpha opportunities could grow as stock dispersion improves from recent lows

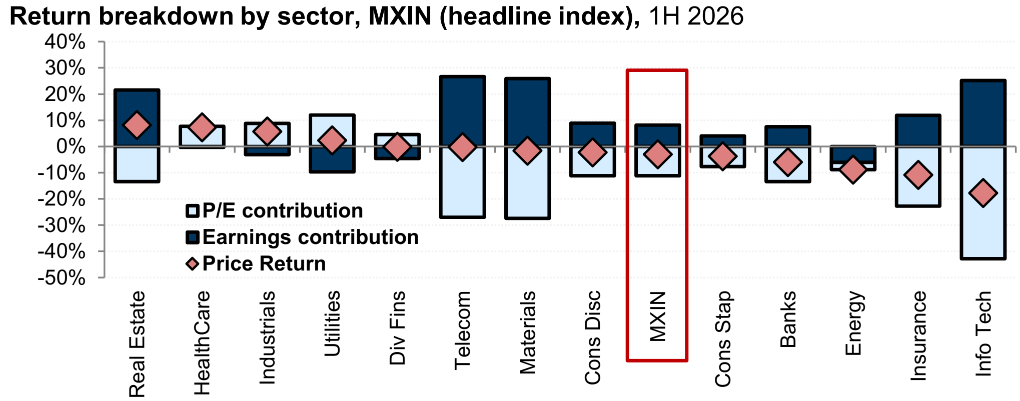


Source: FactSet, Goldman Sachs Global Investment Research

4. Rotation to 'Value' from 'Growth'

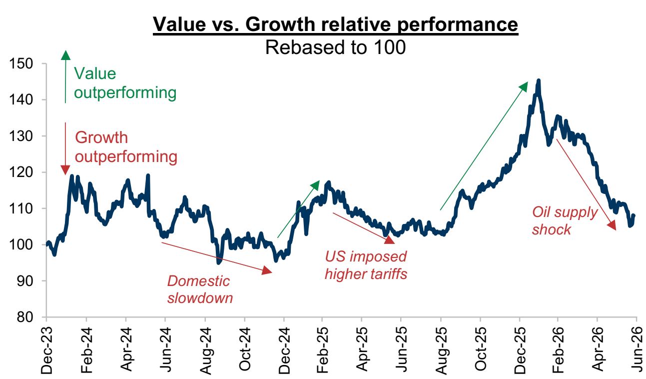
MSCI India declined by 5% in 1H 2026, characterised by 10% de-rating in valuation amid fears of economic slowdown, while forward earnings caught up. Stylistically, 'Growth' factor outperformed 'Value' factor as investors looked for growth that was scarce in a slowdown environment. Looking ahead, we expect a rotation from 'Growth' to 'Value' as investors look for 'undervalued' or 'reasonably valued' pockets in anticipation of a recovery ahead. Breadth of stocks trading at reasonable multiples has improved moderately to 2-3 year highs, providing an opportunity to pick reasonably valued stocks.

Exhibit 11: 1H was characterized by valuation de-rating in several parts of the market



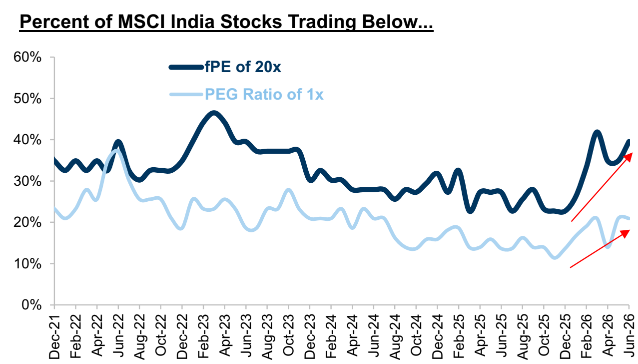
Source: FactSet, MSCI, Goldman Sachs Global Investment Research

Exhibit 12: Stylistically, Value tends to outperform right after a negative market shock; Value has sharply underperformed amid the oil-shock and could see some rotation as investors look for 'value' post the recent correction



Source: Goldman Sachs Global Investment Research

Exhibit 13: Breadth of stocks trading at reasonable multiples has improved moderately to 2-3 year highs

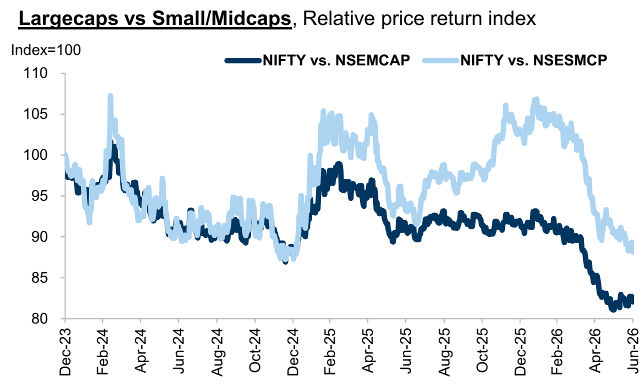


Source: Goldman Sachs Global Investment Research

5. Size Rotation to Large-caps From Mid-caps

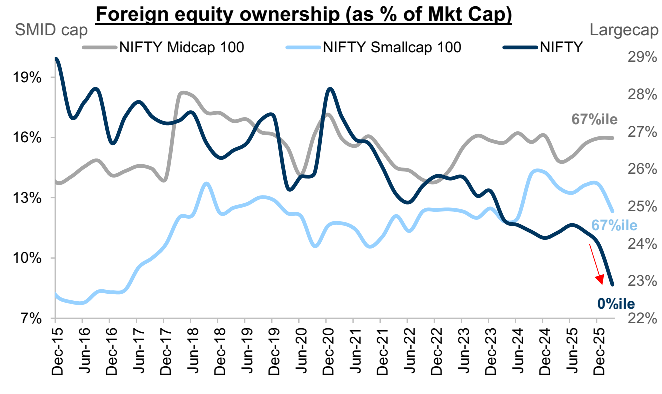
Large-caps could see a tactical rotation on improving foreign sentiment and more reasonable valuations. Large-cap NIFTY has underperformed NSE Mid-cap and Small-cap indices by 10-15pp in 1H, driven by significant foreign selling pressure. **Foreign ownership** in Large-caps dropped to decade-low levels in Q1, while remaining stable in Mid-caps, suggesting Large-caps faced the largest foreign selling pressure. As foreign sentiment improves ahead, we could see some rotation back into Large-caps. **Valuations** for Large-caps now look more palatable, having de-rated close to their 15Y average multiples, even as Mid-caps still trade significantly (+1.5sd) above average. Large-caps currently trade at a historically wide discount of 30% to Mid-caps. Growth-adjusted valuations (PEG ratio) for Large-caps also look more reasonable now as their PEG ratio differential with Mid-caps has narrowed to 2020 lows. There are decent stock-picking opportunities in Large-caps with a wider breadth of 'lower' valuation stocks. **Fundamentally**, Large-caps have seen relatively shallower earnings cuts than the Mid-caps in 1H, and provide better earnings visibility.

Exhibit 14: Large-caps could see a tactical rotation after sharp underperformance against Small/Mid-caps in 1H



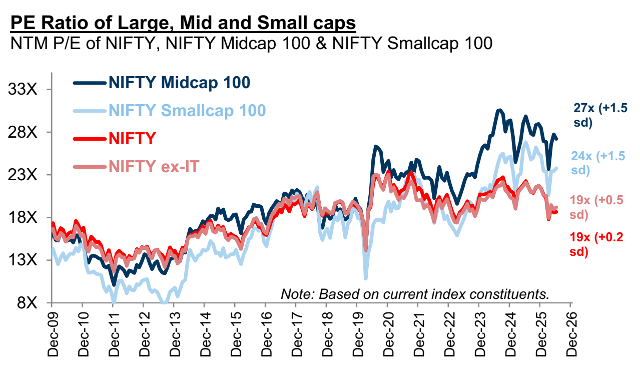
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 15: Large-caps could benefit tactically from abating selling pressure from FIIs, given FII selling YTD was largely concentrated in Large-caps



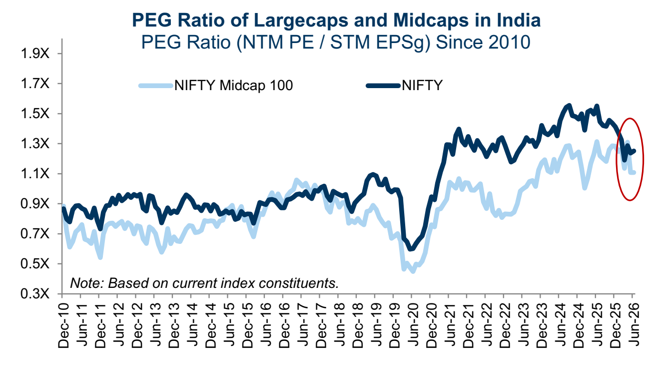
Source: Goldman Sachs Global Investment Research

Exhibit 16: Large-caps have de-rated close to their 15Y average multiples, while Small/Mid-caps still trade at elevated multiples



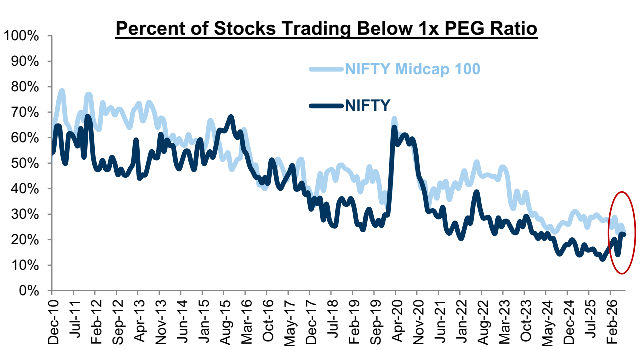
Source: Goldman Sachs Global Investment Research

Exhibit 17: Large-cap growth-adjusted valuations (PEG ratio) are now close to Mid-caps for the first time in many years



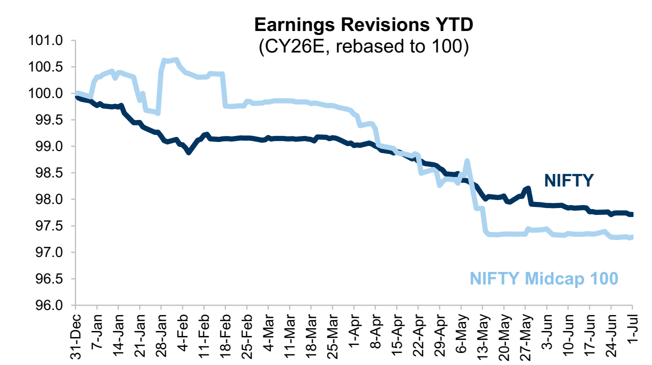
Source: Goldman Sachs Global Investment Research

Exhibit 18: Breadth of low PEG ratio stocks in Large-caps looks as good as Mid-caps for the first time in over 5 years



Source: Goldman Sachs Global Investment Research

Exhibit 19: Large-caps so far have seen relatively lower earnings cuts than the Mid-caps

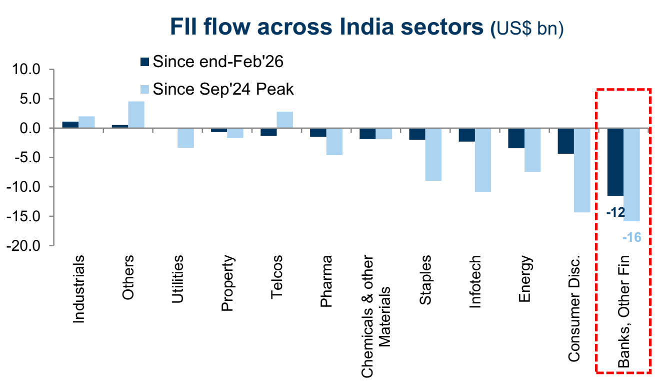


Source: FactSet, Goldman Sachs Global Investment Research

6. Banks Likely to Outperform On Supportive Positioning and Policy Tailwinds in 2H

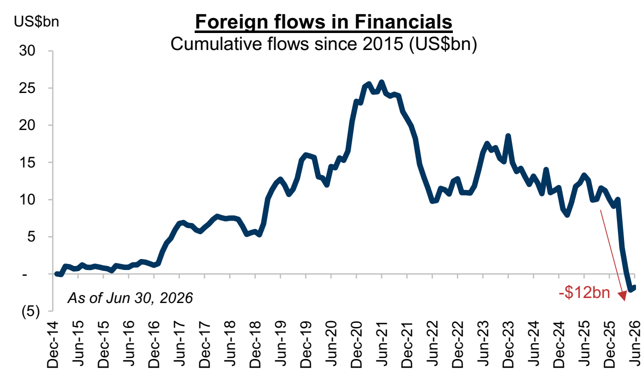
Foreign positioning looks supportive in India Financials. Global investors sold US$12bn of Financials in just the last 4 months, effectively resetting cumulative decade-long inflows to zero. As foreign sentiment turns, Financials stand to benefit the most. **Valuations look the most reasonable** in Financials compared to the rest of the market on various metrics: (1) absolute forward PE is 16x, much lower than the 20x multiple for broad MSCI India index, (2) growth-adjusted PE (PEG ratio) is just under 1x, the only sector in India below the 1x mark, and (3) the sector is trading at the lower end of its own historical range, about 1 s.d. below its 10-year average. Within Financials, Banks specifically look well-placed in the current environment of strong credit growth, **improved liquidity resulting from RBI's recent capital flow measures** (our economists estimate c.US$60bn bond market inflows in CY26), and healthy asset quality. A likely 'super' El Niño could be a cause of concern around asset quality for NBFCs, underpinning our preference for Banks over NBFCs. Banks have outperformed the MSCI India index since RBI's capital flow measures in early June, but still remain below past peaks.

Exhibit 20: Financials will be the biggest beneficiary of easing pressure from foreign selling



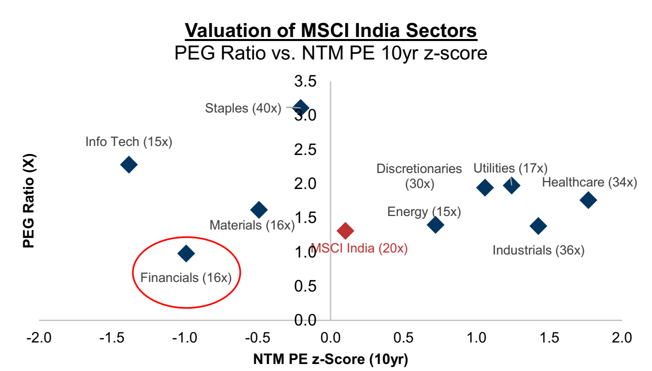
Source: NSDL, Goldman Sachs Global Investment Research

Exhibit 21: Foreigners sold US$12bn in just the last 4 months, effectively resetting cumulative decade-long inflows to zero



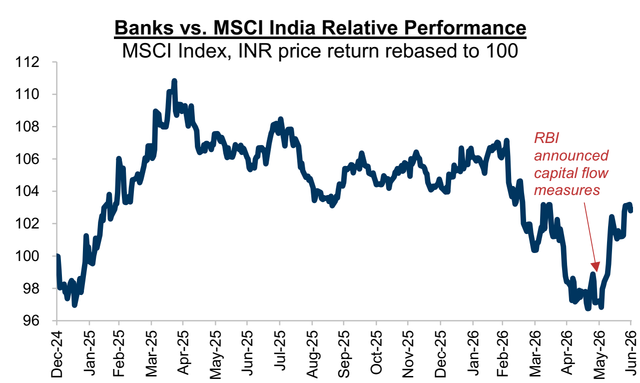
Source: NSDL, Goldman Sachs Global Investment Research

Exhibit 22: Financials screen as the most reasonably valued sector based on absolute PE, growth-adjusted PE and the historical range



Source: Goldman Sachs Global Investment Research

Exhibit 23: Banks have outperformed the market since the RBI announced various capital flow measures in early June, but still remain below past peaks

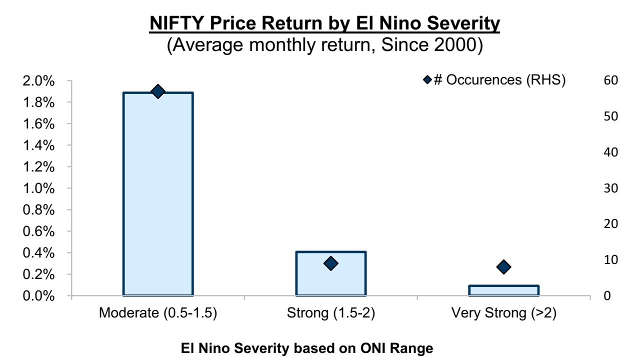


Source: FactSet, MSCI, Goldman Sachs Global Investment Research

7. Power To Outperform Agriculture In A Strong El Niño Event

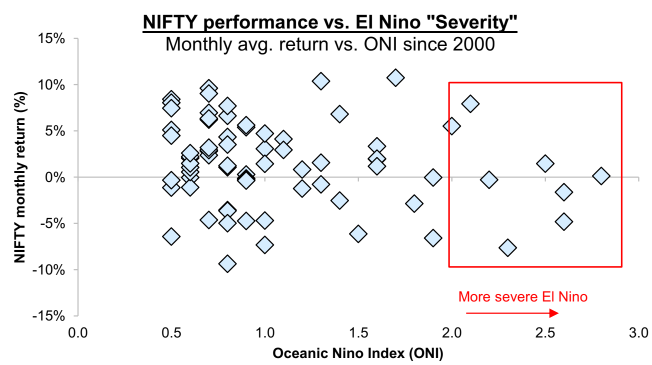
A strong or a very strong ('super') El Niño is likely a headwind for Indian equities overall, but will potentially create wide sector dispersion. We expect Power generation companies to outperform, and Agriculture/Rural companies to underperform amid concerns around a drier- and hotter-than normal monsoon season in 2H 2026. National Oceanic and Atmospheric Administration expects El Niño to strengthen into a strong event, with a 63% probability of developing into a very strong ('super') El Niño in Nov-Jan. India Meteorological Department (IMD) expects to see 10% below normal rainfall in 2026 monsoon season, and forecasts temperatures to remain above normal through July, amid El Niño concerns. Power companies are well-placed to benefit from surging electricity demand amid higher temperatures and power supply shortages from constrained hydro-power generation in weaker monsoons. On the other hand, Rural-exposed companies and Agriculture-oriented businesses like fertilizers/seeds, agri machinery, irrigation, food storage and processing could underperform near-term in a weaker monsoon.

Exhibit 24: Indian equity returns in the past 25 years have been weaker during strong and super El Niño periods



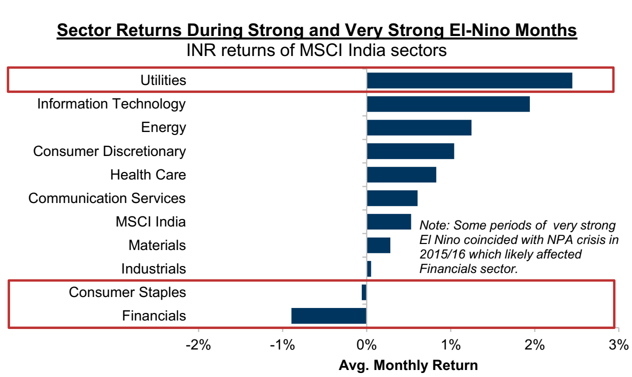
Source: US National Oceanic and Atmospheric Administration, FactSet, Goldman Sachs Global Investment Research

Exhibit 25: Monthly returns have been largely negative in very strong El Niño months



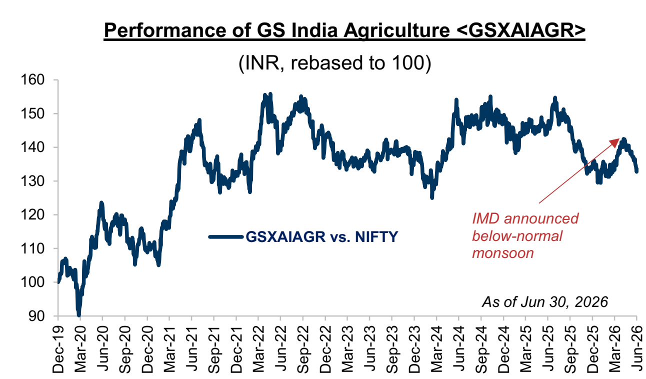
Source: US National Oceanic and Atmospheric Administration, FactSet, Goldman Sachs Global Investment Research

Exhibit 26: Power (utilities) has been the largest beneficiary of strong El Niño periods in the past, while Staples have been more vulnerable



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 27: Agriculture, which includes companies exposed to agri inputs, machinery and infra, has been an underperformer since onset of weaker monsoon in 2026

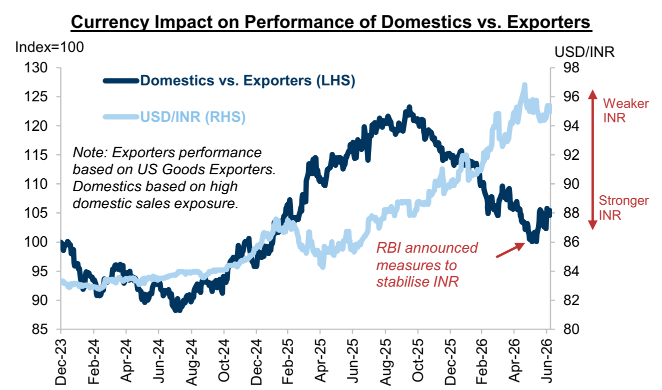


Source: Goldman Sachs Global Investment Research

8. Domestics to Outperform Exporters Amid Stabilizing INR

INR came under significant pressure between end-February and late May, with USD/INR briefly breaking above 96 before easing back below 95 in early June. However, our economists have turned more constructive on INR following recent RBI's capital flow measures and lower oil prices, and expect stronger capital inflows through bond market, both supportive for domestic profitability and growth. Goods exporters were one of the best performing stocks in the market in 1H, supported by a combination of trade deal agreements with the US and EU, and a weakening INR. Now that trade deals are likely priced in and INR depreciation is behind, we expect rotation from 'Exporters' to 'Domestics'. Domestics have outperformed Exporters since the RBI's supportive measures for INR, but the pocket still trades close to the trough, providing a good entry point, in our view.

Exhibit 28: Domestic companies are showing early signs of outperformance against Exporters as INR stabilizes, after underperforming in 1H amid lower US tariffs and weaker INR

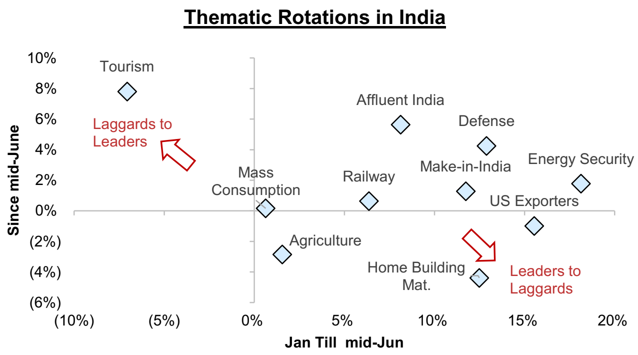


Source: FactSet, Goldman Sachs Global Investment Research

9. Tourism To See Recovery From Lows

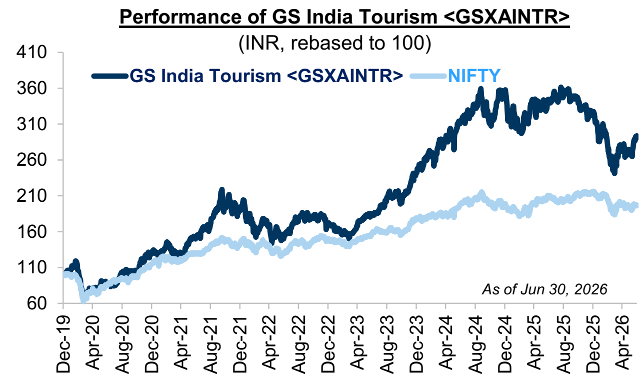
Tourism is likely to see potential earnings upgrades ahead, amid abating cost headwinds for the airline sector, more resilient than anticipated domestic demand for hotels, and likely improvement in travel demand in 2H FY27. Investors have started pricing in some of this recovery as Tourism, the weakest pocket through 1H 2026, has turned into the best performing theme in the recent couple of weeks. Despite the recent outperformance, the theme is still significantly below prior peaks and hence has further runway to grow.

Exhibit 29: Tourism has turned into a leading theme recently, after lagging through most of 1H



Source: Goldman Sachs Global Investment Research

Exhibit 30: Tourism companies are still below prior peaks and have further runway for outperformance



Source: FactSet, Bloomberg, Goldman Sachs Global Investment Research

10. Trade Ideas For 2H

Thematically, we recommend going long on banks, tourism, energy refiners and defense, and favor Large-caps over Mid-caps, value over growth, power utilities over rural/agriculture stocks, and domestics over exporters.

Structurally, we remain bullish on defense and energy security on being strategically important priorities for the government in the medium-term.

Sectorally, we raise utilities to overweight on potential power shortages, and a likely stylistic rotation from growth to value. We remain overweight on banks, energy refiners, TMT, and defense. We raise **industrials** to marketweight on potential recovery in travel/aviation, coupled with reasonable growth-adjusted valuations supported by sustained growth from data centre infrastructure demand. We neutralize **staples** to marketweight on two-sided risks: the sector is seeing strong demand pickup and a relatively insulated earnings impact from input cost inflation, but has risk to de-rate on concerns around strong El Niño. We lower **metals & mining and cement** to underweight on seasonal weakness during monsoon, and lingering cost pressures from commodity inflation and a weaker INR yoy. We stay underweight exporters (IT and pharma) and downstream oil (OMCs).

For **stock-picks**, we highlight 15 large-cap stocks that are reasonably valued, and could benefit from turning foreign sentiment, near-term macro catalysts (like 2Q results, strong bond inflows, and super El Nino), or structural growth themes (like defense and energy security).

Exhibit 31: We raise utilities to overweight, and industrials to marketweight; We lower metals & mining and cement to underweight and staples to marketweight

India Sector Allocations		
Overweight	**Marketweight**	**Underweight**
Banks	Div. Fins (Insurance, NBFCs)	Infotech
Energy (Refiners, E&P)	Industrials (ex-Defense) ↑	Pharma
TMT (Telcos, Internet)	Consumer Discretionary (Autos, Cons. Retail/Services)	Metals & Mining ↓
Utilities ↑	Staples ↓	Cement ↓
Defense	Durables	Chemicals/Materials
	Real Estate	Oil Marketing Companies

Source: Goldman Sachs Global Investment Research

Exhibit 32: We identify 15 large-cap stocks that could benefit from key themes and catalysts in 2H

Bottom Quartile, or Below Sector, or <0

			Size and Liquidity		Growth & Valuations				Rotation Themes			Tactical Themes			Structural Themes	
BBG Ticker	Company Name	Sector	Listed Market Cap (US$bn)	6M ADVT (US$mn)	2026/27 EPS Growth (%)	NTM P/E (x)	PEG Ratio (x)	NTM PE z-score (10y hist.)	Large Cap	Foreign Rotation	Reasonable Valuation	Super El Niño	Recovery Post Oil Shock	INR Stabiliz-ation	Aerospace & Defense	Energy Security
Thematic Plays for 2H																
RELIANCE IS	Reliance Industries	Energy	187	263	9	20x	2.3x	0.1	✔		✔		✔			✔
HDFCB IS	HDFC Bank	Financials	134	366	13	15x	1.1x	-1.8	✔	✔	✔			✔		
ADANI IS	Adani Power	Utilities	44	78	9	30x	2.0x	-0.3	✔		✔	✔				
ADE IS	Adani Enterprises	Industrials	44	63	215	70x	1.1x	0.5	✔		✔					✔
KMB IS	Kotak Mahindra Bank	Financials	40	82	16	22x	1.4x	-1.5	✔		✔			✔		
NTPC IS	NTPC Corp.	Utilities	36	52	9	14x	2.9x	0.9	✔		✔	✔				✔
HNAL IS	Hindustan Aeronautics	Industrials	31	68	10	29x	2.8x	0.6	✔	✔	✔				✔	
ETERNAL IS	Eternal (Zomato)	Discretionary	29	137	184	108x	1.5x	-0.8	✔	✔	✔		✔	✔		
PWGR IS	Power Grid of India	Utilities	28	47	7	15x	2.3x	1.1	✔	✔	✔	✔				✔
ADANIGR IS	Adani Green Energy	Utilities	27	53	71	70x	1.4x	-0.6	✔		✔	✔				✔
INDIGO IS	InterGlobe Aviation	Industrials	22	75	131	37x	0.5x	0.2	✔	✔	✔		✔			
HDFCLIFE IS	HDFC Life Insurance	Financials	13	30	16	53x	3.8x	-1.8	✔	✔	✔			✔		
IH IS	Indian Hotels	Discretionary	11	21	18	45x	2.8x	-0.2	✔		✔		✔			
MAZDOCKS IS	Mazagon Dock Shipb.	Industrials	11	45	8	34x	5.4x	0.9	✔		✔				✔	
MMYT UW	MakeMyTrip	Discretionary	5	82	14	91x	1.9x	-0.2	✔	✔	✔		✔			
Median			**29**	**68**	**14**	**34x**	**2.0x**	**-0.2**								

Note: (1) Based on stocks within BSE200 and GS coverage universe. (2) Largecap exposure based on market cap size or inclusion in NSE100 index; (3) Foreign Rotation Candidates based on YTD reduction in OW/UW exposure by foreign funds; (4) Reasonable valuations determined based on historical and relative PE ratio and PEG ratios; (5) Super En Niño beneficiaries include power utility companies and exclude Agri / Rural companies; (6) Oil shock recovery candidates include tourism, consumer retail/services and energy refiners; (7) INR stabilisation candidates include domestic companies with stable growth and excludes exporters; (8) Defense and Energy Security companies identified based on our keyword analysis published previously. (9) Valuations and earnings growth forecasts are based on I/B/E/S and FactSet estimates. (10) Pricing as of Jul 07, 2026.

Source: FactSet, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Amorita Goel, CFA, Sunil Koul and Timothy Moe, CFA, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Amorita Goel, CFA Goldman Sachs (Singapore) Pte, Sunil Koul Goldman Sachs International, Timothy Moe, CFA Goldman Sachs (Singapore) Pte.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Other Disclosures

Marquee is a product of the Goldman Sachs Global Markets Division. Any Marquee content linked in this report is not necessarily representative of GS Research views.

All MSCI data used in this report is the exclusive property of MSCI, Inc. (MSCI). Without prior written permission of MSCI, this information and any other MSCI intellectual property may not be reproduced or redisseminated in any form and may not be used to create any financial instruments or products or any indices. This information is provided on an "as is" basis, and the user of this information assumes the entire risk of any use made of this information. Neither MSCI, any of its affiliates nor any third party involved in, or related to, computing or compiling the data makes any express or implied warranties or representations with respect to this information (or the results to be obtained by the use thereof), and MSCI, its affiliates and any such third party hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any of this information. Without limiting any of the foregoing, in no event shall MSCI, any of its affiliates or any third party involved in, or related to, computing or compiling the data have any liability for any direct, indirect, special, punitive, consequential or any other damages (including lost profits) even if notified of the possibility of such damages. MSCI and the MSCI indexes are service marks of MSCI and its affiliates. The Global Industry Classification Standard (GICS) were developed by and is the exclusive property of MSCI and Standard & Poor's. GICS is a service mark of MSCI and S&P and has been licensed for use by The Goldman Sachs Group, Inc.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.